MEM 411 METROLOGY HND MANUFACTURING

METROLOGY: Is the Science of Weights and Measures or of Measurement. (A System of Weights and Measures.

MEASUREMENT:- Is the act of Measuring, it can also be a Magnitude or amount ~~of~~ determined by an act of Measuring.

The bases of the Metric System of length is the International Prototype Standard Metre, which is defined as the distance between two scratches on a platinum-Iridium bar. The Prototype Metre is kept under Special Conditions in the Achieves of the International Bureau of Weights and Measurements.

The Measuring Instruments/Methods can be Classified in Various Manners as given below.

1a DIRECT MEASURING: In this, the Measured Value is determined directly eg Micrometer, Vernier caliper, Vernier height gauge, Bevel Protractor etc such Instruments are Simple and most widely used in Production.

1b INDIRECT MEASURING:- In this, the dimension is determined by Measuring other Values ~~of~~ ~~the~~ functionally related to the required Value eg divider, caliper, Sine bar etc.

2a. ABSOLUTE: Here, the Zero division of the Instrument Corresponds with the zero Value of the Measured dimension eg Steel rules, Vernier Caliper

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2b. COMPARATIVE: Here, only the deviations of the measured dimensions from a master gauge are determined eg dial indicators, optimeters

3a. CONTACT: Here, the measuring tip of the instrument actually touches the surface to be measured eg micrometers, calipers, dial indicators.

3b CONTACTLESS:- Here, no contact is required for measurement eg Tool Maker's microscope, Projection Comparator.

4 According to their functions, the measuring instruments are classified as.

(a) LINEAR OR LENGTH MEASURING INSTRUMENT

Instruments used for measuring linear or length are, steel rule, caliper, divider, micrometer, vernier caliper etc.

(b) ANGULAR OR ANGLE MEASURING INSTRUMENT

The instrument used for measuring angular or angles are:- Combination set, bevel protractor, sine bar, square, dividing head.

(5) Depending on their accuracy, the measuring instruments can be grouped into three (3) categories:

(i) Most accurate group includes light interference instruments.

ERRORS IN ENGINEERING MEASUREMENT.

An error may be defined as the difference between the measured and actual values.
Example, If the two operators use the same device or instrument for measurement, it is not necessary ~~for~~ that both operators get similar result. The difference between the measurement is referred to as an Error.

⇒ TYPES OF ERRORS COMMONLY FOUND IN ENGG MEASUREMENT.

The errors that may occur in the measurement of a physical quantity can be classified into 6 types which includes:

1. Constant error
2. Systematic error
3. Random error
4. Absolute error
5. Relative error
6. Percentage error.

⇒ SOURCES OF ERRORS IN MEASUREMENT.

Common sources of error include:

1. Instrumental error
2. Environmental error
3. Procedural error
4. Human error.

All of these errors can be either be Random or Systematic, depending on how they effect the result.

CLASSES OF ERROR.

There are two major classes of errors in a measurement (1) Limitations in the sensitivity of the instrument used.

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(2) Imperfections in the techniques used to Make the Measurement.

⇒ EQUIPMENT ERROS:
Equipment error Means a reproducible failure of the equipment to operate in accordance with such equipment's Published specifications.

⇒ CAUSES OF EQUIPMENT FAILURE:
Physical Wear and Tear: This causes of Industrial Machine or equipment failure Includes
(1) Bearing failure (2) Metal fatique (3) Corrosion
(4) Misaligment (5) General Surface degradation.

⇒ EXAMPLE OF EQUIPMENT FAILURE.
Engine failure or Misfire, break failure or stop Controlling device failure, Suspension of operation due to heat.

⇒ OPERATIONAL INTERFERENCE.
A Machine or Piece of equipment that is operational is in uce or is ready for use in circumstance that interfere Materially and Substantially with the orderly operation of the equipment. Also in determing what Conditions would Make it Impossible or un-ressonably hazardous to carry on the operation of the Machine.

(ii) Second group includes:- Optimeters, tool-Maker's Microscope and dial Comparators.
(iii) Third group includes:- Dial indicators, Vernier Calipers, Micrometers and rules with Vernier scales.

6. Measuring Instruments can also be classified in accordance with their meterological properties, such as range of measurement, scale graduation value, scale spacing, sensitivity and reading accuracy.

The various terms connected with the measuring instruments can be defined as given below:-

1. Range of Measurement:- It indicates the size values between which measurements can be made on the given instrument eg micrometers are usually available for the following size ranges 0 to 25, 25 to 50, 50 to 75, 75 to 100, 100 to 125 and 125 to 150 mm.

(ii) Scale division Value: It is the measured value corresponding to one division of the instrument scale eg The scale division value of a micrometer is 0.01mm.

(iii) Scale spacing:- It's merely the distance between the axes of two adjacent graduations on the scale.

(iv) Sensitivity: The sensitivity of an instrument is the ratio of the scale spacing to the scale division value.

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If on a dial Indicator, the scale spacing is 1.5mm and the scale division value is 0.01mm then sensitivity is 150. Sensitivity is also called "amplification factor" or 'gearing ratio'.

The reading accuracy of an Instrument depends to a large extent on the contact pressure. To prevent error due to variation of contact pressure special devices, which automatically adjust the contact pressure are incorporated in the design of measuring instrument eg the ratchet stop of a micrometer.

The measuring accuracy also depends upon variation in temperature during measuring. Due to this, when taking measurements, the job and the instrument must be at the same temperature.

TYPES OF LENGTH STANDARD:

A length may be expressed as the distance b/w two lines or as the distance b/w two faces. So the instruments used for the direct measurement of linear dimensions fall into two categories:

1. Line standards (ii) End standard.

(i) Line standards: Here, the measurement is made b/w two parallel lines engraved across the standard. The most common example of line standard or line measurement is the rule, with its division shown as lines marked on it.

(ii) End Standards:- Here, the measurement is made b/w two flats parallel faces eg slip gauges end bars, micrometer, vernier calipers etc.

NB Line standards have the drawback that the engraved line themselves posses thickness. Also, the assistance of a magnifying glass or a microscope is required if sufficient accuracy is to be obtained. Greater accuracy can be obtained with the use of end standards as compared to line standards. Due to this, end standards are employed in the shop as far as possible.

INSTALLATION $ TROUBLESHOOTING.

It is a systermatic approach to problem solving that is often used to find and correct issues with complex machines, electronics, computers $ software systems.

⇒ WHAT IS A DRUNKEN THREAD:

This is the one having erratic pitch in which the advance of the helix is irregular in one complete revelution of the ~~fre~~ thread.

⇒ WHAT ARE THE DIFFERENT TYPES OF ERROR IN THREAD?

The differentetypes of error in thread include.
1. Major diameter 2. Minor diameter 3. Effective diameter 4. pitch 5. flank angles 6. crest and root errors.

⇒ EFFECT OF A DRUNKEN THREAD.

Thread roots with drunken threads have an erratic pitch, creating irregularities in revolutions.

LIMITS, TOLERANCE AND FITS.

Limits is a value to which a sequence converges. Equivalently the common value of the upper limit and the lower limit of a sequence.

<u>NB</u> If the upper and lower limits are different, then the sequence has no limit (ie does not converge.)

TERMINOLOGY FOR LIMITS AND FITS

1. The type of fit which is required b/w the mating parts.
2. The tolerance which is to be allowed each dimension.

⇒ Type of fit and its terminology.

⇒ Shaft:- The term shaft refers not only to the diameter of a circular shaft, but also to any external dimension on a component.

⇒ Hole:- The term hole refers not only to the diameter of a circular hole, but also to any internal dimension on a component.

When an assembly is made of two parts one is known as Male (enveloped)

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Surface, and the other one is famale (enveloping) Surface. The male Surface is referred as 'shaft' and the famale Surface as 'Hole'.

⇒ Basic Size:- Basic Size or normal Size is the Standard Size for the part and is the same both for the hole and its Shaft. This is the Size which is obtained by Calculation for strength.

⇒ Actual Size:- Actual Size is the dimension as measured on a manufactured part. As already noted, the actual Size will never be equal to the basic Size and it is sufficient, If it is within predeter mined limits.

⇒ Limits Size:- These are the Max and Min permissible Sizes of the part.

⇒ Maximum Limit:- Max limit or High limit is the Max Size permitted for the part.

⇒ Minimum Limit:- Minimum limit or Lower limit is the minimum Size permitted for the part.

⇒ Tolerance:- Tolerance is the differente b/w max limit size and Minimum limit Size.

⇒ Allowance:- Allowance is an International difference b/w the Max Material limits of mating part.
For shaft, the Max Material limit will be it high limit and
for hole, it will be its low limit.

<u>NB</u> If the shaft is smaller than hole the allowance is positive (+ve) but if the shaft is larger than the hole it is negative (−). The above defination are explained in the figure below.

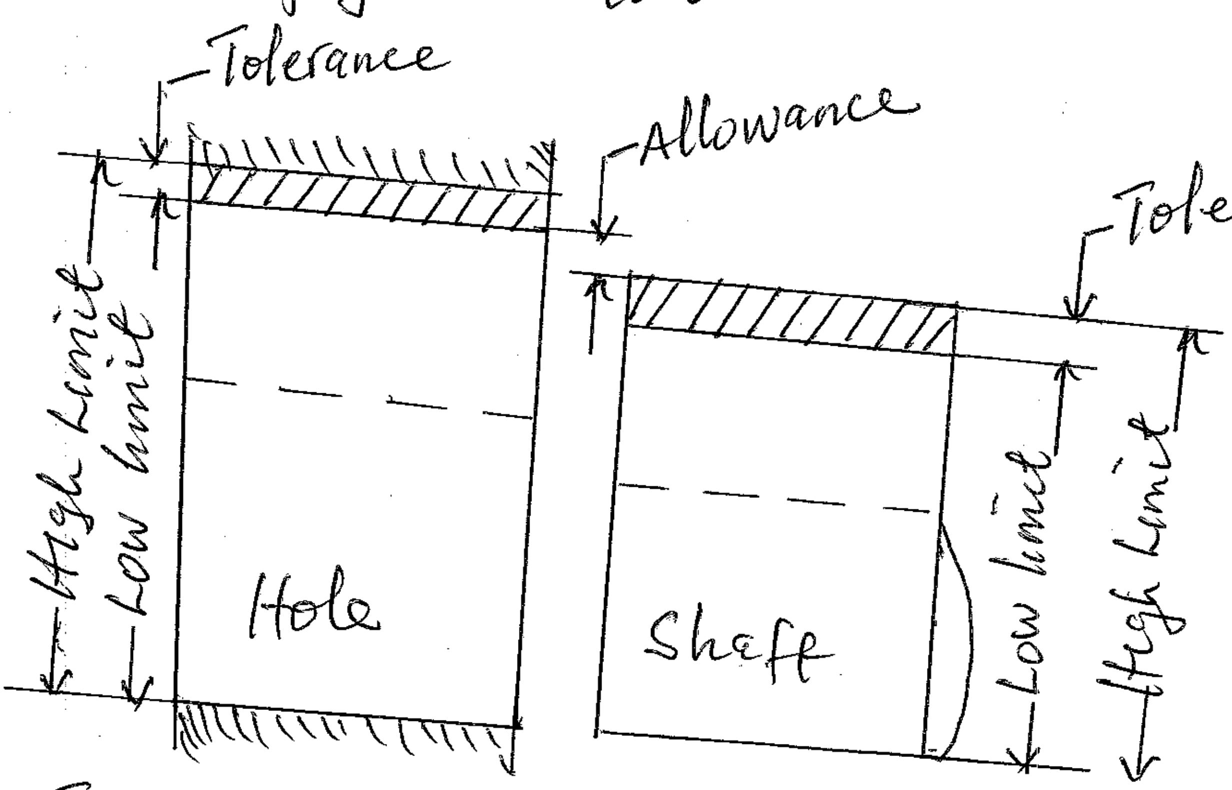

Fig 1. Limits and Tolerance

⇒ Deviation: It is the algebraic difference b/w a size (actual, Max, minimum etc) and the corresponding basic size.

Actual Deviation: It is the algebraic difference b/w an actual size and the corresponding basic size.

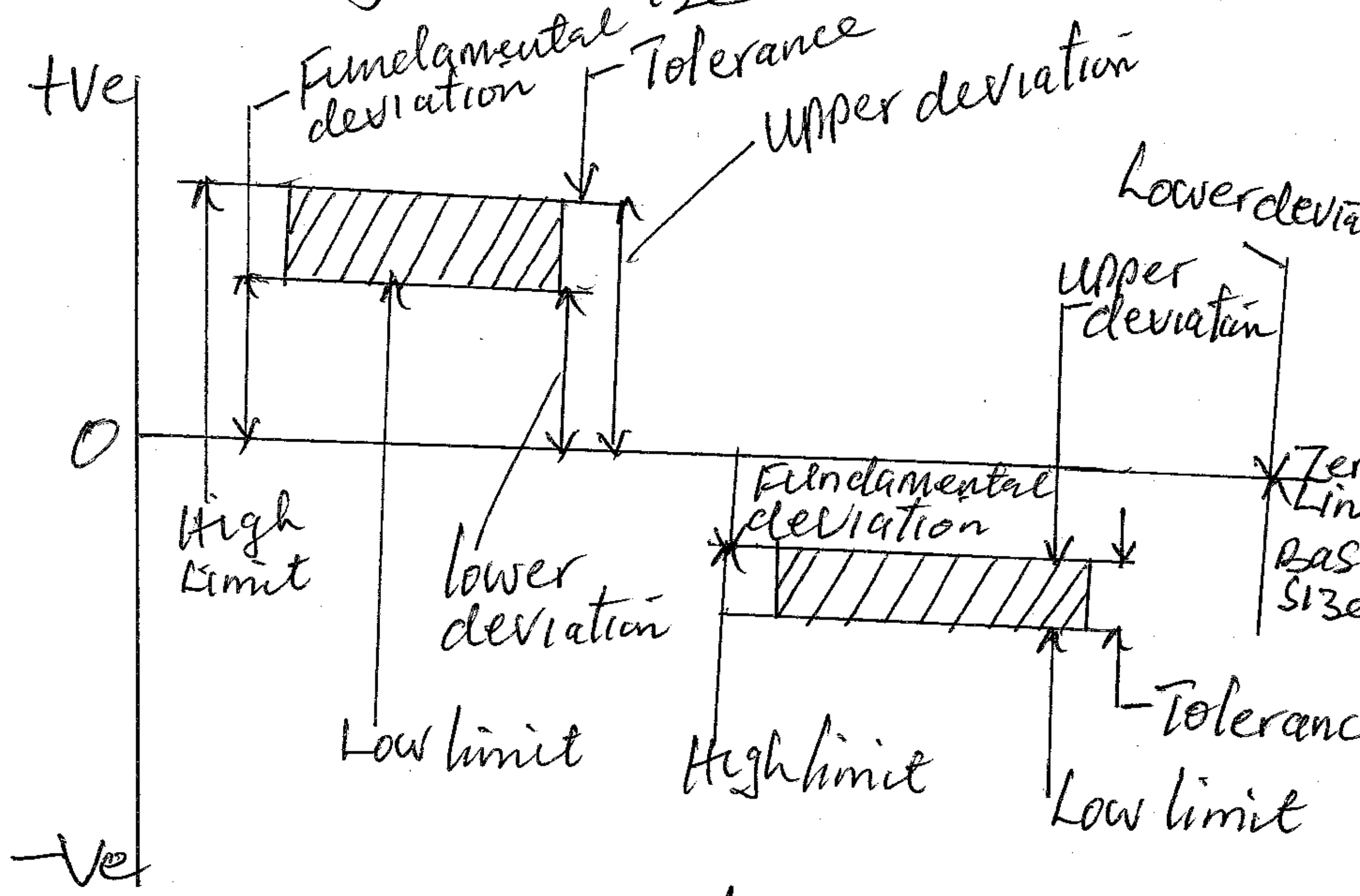

Fig 2. Deviations.

⇒ Upper Deviation: It is the algebraic difference b/w the Max limit of size and the corresponding basic size. It is a positive ~~quatity~~ quantity when the Max limit of size is greater than the basic size and a negative quantity when the Max limit of size is less than the basic size.

⇒ Lower Deviation:- It is the algebraic difference b/w the minimum limit of size and corresponding basic size. It is positive (+ve) quantity when the minimum limit of size is greater than the basic size. It's a −ve quantity when the minimum limit of size is less than the basic size.

⇒ Mean Deviation:- It is the arithmatic mean deviation b/w ~~the~~ upper deviation and lower deviation.

⇒ Zero line:- It is a straight line to which the deviation are referred to in a graphical presentation of limits and fits. It is a line of zero deviation and represent the basic size. When the zero line is drawn horizontally positive (+ve) deviation are shown and the (−ve) deviation below this line.

⇒ Fundamental Deviation: This is the deviation either the upper or the lower deviation, which is the nearer one to the zero line for either a hole or a shaft, it fixes the position of the tolerance zone in relation to the zero line.

⇒ Tolerance zone:- It is the zone bounded by the two limits of size of a part in the graphical representation of tolerance. It's define by it magnitude and by its position in relation to the zero line.

⇒ Unilateral Limits:- In the method of presenting the limits both the limits of size are on the same side of the zero line. That is the permitted tolerance is stated or indicated as wholly +ve or wholly −ve.

eg $30\,mm^{+0.13}_{+0.00}$ or $100\,mm^{-0.12}_{-0.26}$ one of limits of the size may be the basic size.

⇒ Bilateral Limits:- Here, one of the limits of size on one side of the zero line and the othe limit of size is on the other side of the zero line. ie the permitted tolerance is indicated partly +ve and partly −ve

eg $90\,mm^{+0.010}_{-0.025}$

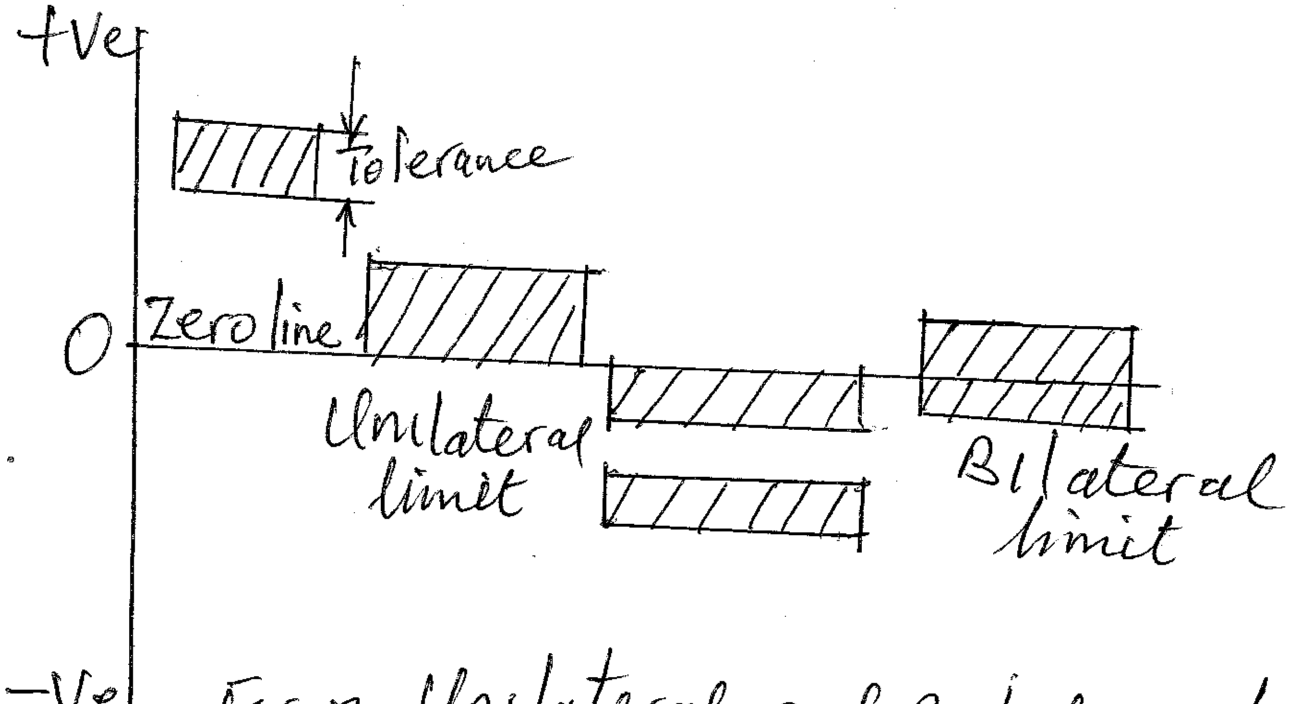

Fig 3. Unilateral and Bilateral limits

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FITS:- The fit between two mating parts is the relationship existing b/w them with respect to the amount of play or Interference which is present when they are assembled together. According the fit may result either in movable joint or a permanent joint: eg. a shaft running in a bush can move in relation to it and so form a moving joint, whereas a pulley mounted on the shaft forms a fixed joint. The nature of joint or fit is characterised by the presence and size of clearance, (for movable joints) or Interference (for fixed joint).

⇒ Basic types of fits:-
There are three basic types of fits

1. Clearance fit:- In clearance fit or running fit the shaft is always smaller than hole. A positive allowance exists b/w the largest possible shaft and the smallest possible hole ie when the shaft and hole are at their Max Metal Conditions. The tolerance zone of the hole is entirely above that of the shaft.

⇒ Minimum clearance:- It is the difference b/w the max size of shaft and Minimum size of hole.

⇒ Maximum clearance:- It is the difference b/w the minimum size of shaft and Maximum size of hole.

2. Interference, Press or force fit:-
In this type of fit, the shaft is always largest than the hole. The tolerance zone of the shaft is entirely above that of the hole.

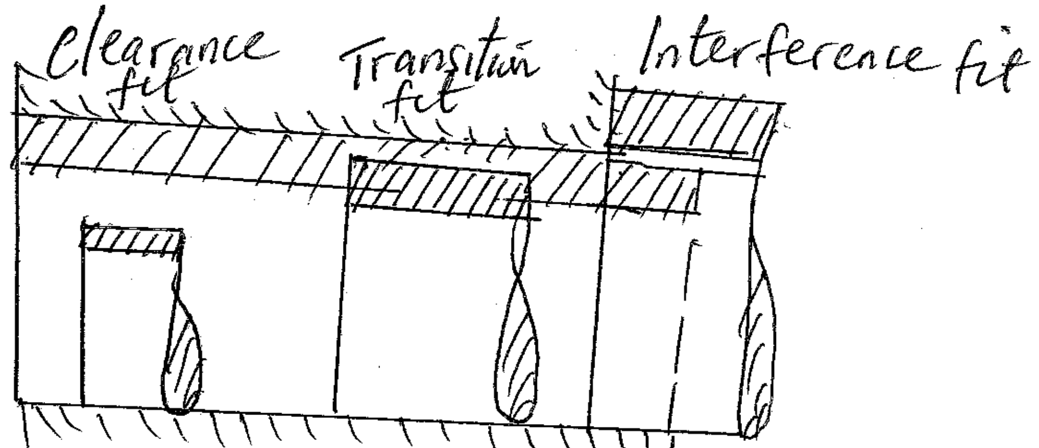
⇒ Minimum Interference:- is the difference b/w the max size of hole and the minimum size of ~~the~~ shaft prior to assembly.

⇒ Maximum Interference:- It is the difference b/w the minimum size of hole and the max. size of shaft prior to assembly.

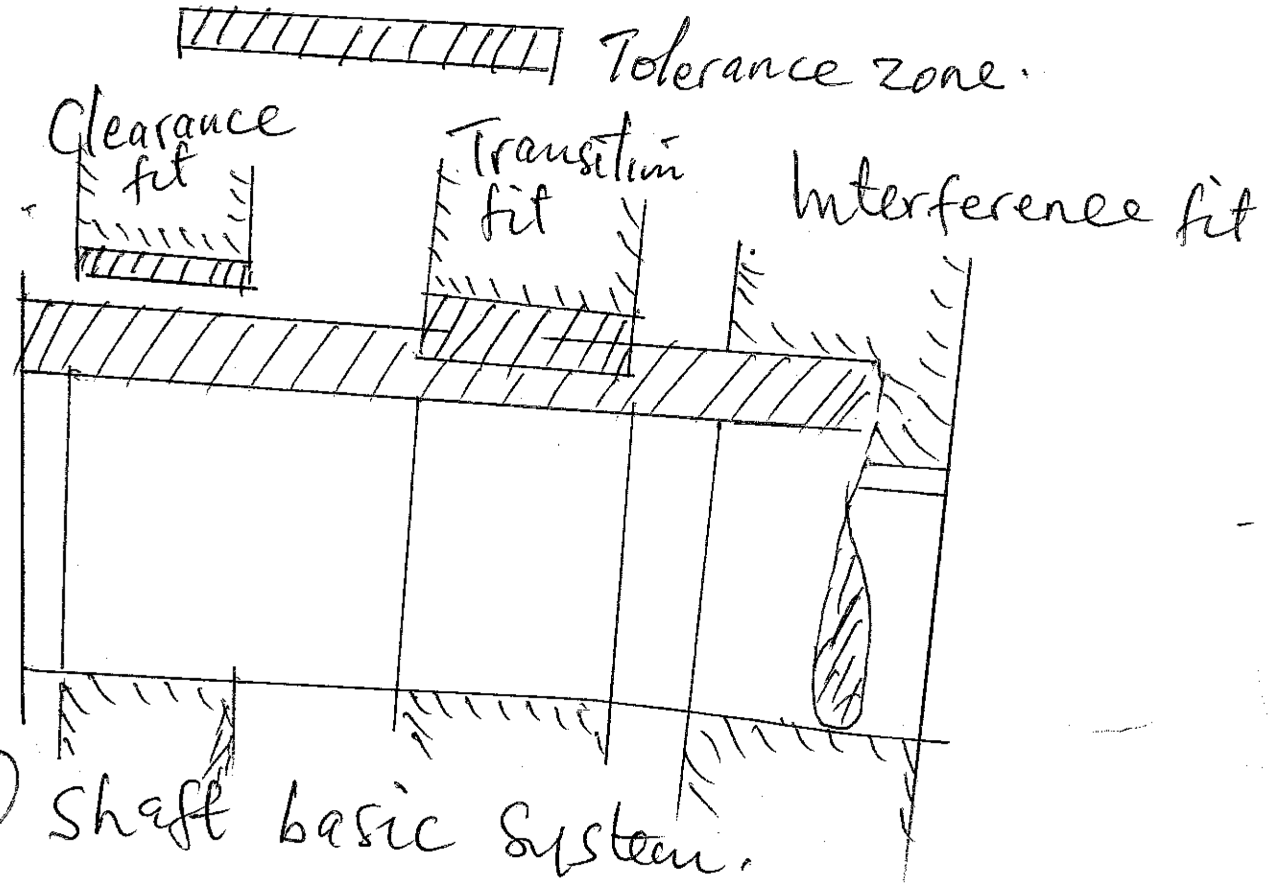
3. TRANSITION OR SLIDING FIT:- It occurs when the resulting fit due to the variations in size of male and female components due to their tolerance, varies b/w clearance and interference fits. The tolerance zones of shaft and hole overlap.

⇒ Hole Basic System:- In this system, the hole is kept constant and the shaft diameter is varied to give the various types of fits. The basic size of the hole is taken as the low limit of size of the hole. That is the maximum metal condition (MMC) of the hole. The high limit of size of the hole and the two limits of size for the shaft are then selected to give the desired fit.

⇒ Shaft Basic System:- Here, the shaft is kept constant and the size of hole is varied to give the various fits. The basic size of the shaft is taken as one of the limits of size (Max limit) for the shaft. That is, its max metal condition (MMC). The other shaft limit of size and the two limits of size for the hole are then selected to give the desired fit.



(a) Hole basic system



(b) Shaft basic system.

NB The most Commonly used fits of clearance type are ① slide fit ② Easy slide ③ running fit ④ Slack running fit ⑤ Loose running fit.

ISO (International Organization for Standardization) System of limits and fits.

The ISO System Covers holes and Shafts from the Smallest size up to 3150 mm. For any size over this range there is a wide choice of fits available and for each of the fits, there is a series of tolerance grades from very fine to wide tolerance. Either a hole based or a shaft based system can be obtained the standard.

For any basic size there are 28 different holes. These are obtained by providing a series of holes which are progressively oversize and a series of hole which are progressively undersize. The difference ~~form~~ from basic size of the various holes is given by the fundamental deviation and it is these differences in size which give the fit required. The 28 holes are designated by the capital letters A, B, C, CD, D, E, EF, F FG, G, H, J, JS, K, M, N, P, R, S, T, U, V, X, Y, Z, ZA, ZB, and ZC. Each of the 28 holes has a choice of 18 grades of tolerance which are designated as IT01, IT0, IT1, to IT 16. The tolerance grades decide the accuracy of manufacture. Similarly for the shafts, there are 28 shafts for a given basic size and these are designated by small letters 'a' to 'zc'. Also each shaft has 18 grades of tolerance which are designated as for the holes.

⇒ Solved Examples.

Example I. Find the values of allowance, hole tolerance and shaft tolerance for the following dimensions of mated parts according to basic hole system. Hole: 37.50 mm, 37.52 mm; shaft 37.47 mm, 37.45 mm

Soln

(i) Hole tolerance = High limit − low limit
= 37.52 − 37.50 mm
= 0.02 mm

(ii) Shaft tolerance = High limit − low limit
= 37.47 − 37.45
= 0.02 mm

(iii) Allowance = Max Metal Condition of Hole − Max Metal Condition of Shaft
= Low limit of hole − High limit of shaft.
= 37.50 − 37.47
= 0.03 mm.

Example 2 A 75mm shaft rotates in a bearing. The tolerance for both shaft and bearing is 0.075mm and the requireed allowance is 0.10mm. Determine the dimensions of the shaft and the bearing bore with the basic hole standard.

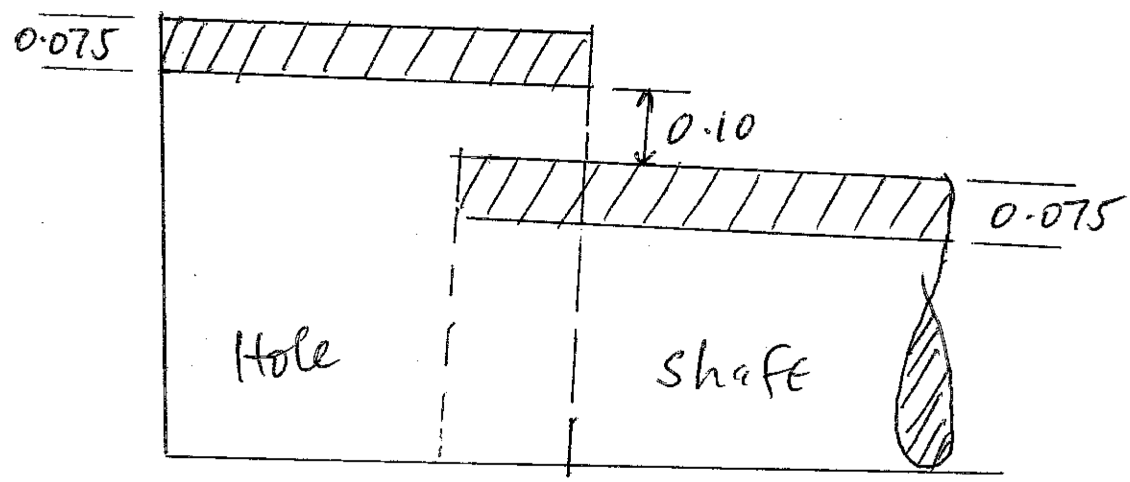
Soln.

Low limit of hole = 75mm

High limit of hole = Low limit + tolerance
= 75 + 0.075
= 75.075mm

High limit of shaft = low limit of hole − allowance
= 75 − 0.10
= 74.90mm

Low limit of shaft = High limit − tolerance
= 74.90 − 0.075
= 74.825mm.



Example 3 In a limit system, the following limits are specified to give a clearance fit between a shaft and a hole.

Shaft $30^{-0.005}_{-0.018}$ mm ϕ hole $30^{+0.020}_{-0.000}$ mm ϕ

Determine (a) Basic size (b) Shaft and hole tolerances (c) The shaft and hole limits (d) The Max and Min. clearance.

<u>Soln</u>.

(a) Basic size = 30mm.

(b) Shaft tolerance = 0.018 − 0.005
= 0.013 mm

Hole tolerance = 0.020 mm

(c) High limit of shaft = 30 − 0.005
= 20.995mm

low limit of shaft = 30 − 0.018
= 29.982 mm

High limit of hole = 30 + 0.020
= 30.020 mm

low limit of hole = 30 mm.

(d) Max clearance = High limit of hole − low limit of shaft
= 30.020 − 29.982
= 0.038mm

Minimum clearance = low limit of hole − High limit of shaft

= 30.000 − 29.995

= 0.005 mm

Example: A 100 mm diameter journal and bearing assembly has a clearance fit, with the following specifications:-

Tolerance on bearing = 0.005 mm

Tolerance on journal = 0.004 mm

Allowance = 0.002 mm.

Determine the size of the bearing and the journal on (i) Hole Basis System
(ii) Shaft Basis System.

Take unilateral system of tolerances.

Soln

(a) Hole Basis System

Lower limit of bearing = Basis size = 100 mm

Highest limit of bearing = L.L of bearing + Tolerance
= 100 + 0.005 = 100.005 mm

Higher limit of journal = Lower limit of bearing − allowance
= 100 − 0.002
= 99.998 mm

Lower limit of journal = Higher limit − Tolerance
= 99.998 − 0.004
= 99.994 mm

(b) Shaft-Basis System

Upper limit of Journal = Basic size = 100mm

Lower limit of Journal = upper limit − Tolerance

= 100 − 0.004

= 99.996 mm

Lower limit of Bearing = upper limit of Journal + allowance

= 100 + 0.002

= 100.002 mm

Upper limit of Bearing = lower limit + tolerance

= 100.002 + 0.005

= 100.007 mm

Class work

For a number of Interchangeable mating parts (holes and shafts), The average allowance is 0.04 mm and the allowance must not exceed ± 0.012 mm from the average value, The basic size is 100mm. Tolerance on hole = 2 × tolerance on the shaft. Determine the sizes of holes and shafts using Hole basis system and Unilateral system of tolerances.

STATISTICAL QUALITY CONTROL AT THE PRODUCTION AND PLANNING.

Introduction:

Quality means the degree of perfection of excellence when applied to manufactured products. It does not necessarily mean 'the best' it should imply the best for the money. That is, the product should meet the desired requirements at lowest cost. The following fact about quality should be noted.

1. The word 'quality' is meaningless unless and until the use of the product is specified. The good quality means that the product serves the purpose well for which it was stipulated.

2. Quality is not absolute but is a relative term, it can only be judged by comparing with some standard. Thus the quality of a unit may be defined as its conformance to standard or specifications.

3. Quality has direct effect on manufacturing cost and on selling price. High quality means high cost and vice versa.

4. Quality is also affected by quantity. The higher the degree of quality tighter becomes controls and more difficult it becomes to achieve quantity output.

By 'quality Control' we means those processes or operations of testing, measuring and comparing the manufactured components with a standard and then determining whether it should be accepted, rejected, adjusted or reworked. It is of course more sensible to control the process so that the standards are being maintained and that defective parts will not be produced, instead of accepting or rejecting a part after it has been manufactured.

In addition, quality Control means the systematic control of those variables in a manufacturing process which affect the excellence of the end-product.

Statistical quality Control (SQC) procedures which are based on well established statistical methods have been developed which provide economical means of maintaining continual analysis and control of processes and products. SQC applies the theory of probability to sample testing or inspection. The inspector on the shop floor takes controlled random samples of parts. These are then checked by measurement or gauging. The quality of the total batch of components is then judged by the result of the sample.

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nat is, if the number of rejects found in the sample is too large, then the whole batch will be rejected and vise-versa. Thus, the two principal objectives of SQC are:-

① To prevent the production of defective parts by inspecting them and establishing control over manufacturing process during production, instead of waiting to inspect them after they have been completed, when it is too late to take corrective action.

② To reduce the cost of inspection and at the same time ensure quality of the product by adopting a sampling inspection plan that is based on statistical methods.

In addition to the above two principal objectives, SQC results in the following benefits.

(i) It results both in decreased scrap and in less time being spent on ~~recti~~ ratification of defective components.

(ii) ~~It result~~ Due to uniformly achieable improved product specifications, the performance of assembly operations may be improved.

(iii). Where inspection has ~~to~~ to be by destrictive testing, the adoption of SQC is the only reasonable guarantee of quality.

(iv) The application of quality control result in investigations which will lea[d to] improved inspection methods, improved ma[nu]facturing methods and improved product specifications.

STATISTICAL TOOLS FOR QUALITY CONTROL

The main statistical tools used in the quality control jobs are:-

1. Frequency distributions
2. Control charts
3. Sampling tables.

⇒ Frequency distribution:-
For any manufacturing method, it is generally recognised that no two parts are exactly alike and also that for every part, there are always variation from established standards. These variations may be large or small. Variation in a product are due to slight differences in successive units of raw material, wear on tools, changed setting on machines, fluctuations in power, changes in temp, and other numerous causes. All these causes or variables can be grouped in to two categories.

1. Chance Variables:- These variables or causes occur at random and they are difficult to avoid. These causes result in variations which are so small that their elimination can

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not be justified economically. These causes are:

(i) Inherent variation in the raw material, that is, materials not entirely homogeneous.

(ii) Imperfections inherent in the design of the machine or in the manufacturing process.

(iii) Natural or inherent inaccuracies of inspection instruments.

(iv) Feel, eyesight or judgement of the production operator.

(2) Assignable Variables: These variables or causes result from:

(i) Trouble in the manufacturing process or improper operation of the machines.

(ii) Incorrect sequence of manufacturing operation

(iii) Machines or inspection instruments worn out and in need of repair.

(iv). Room temps. that vary during the day or in different parts of the same room.

v. Variations of plants building affecting the performance of the machines.

Assignable causes result in large variation in the quality of the product and they should be identified and eliminated, if possible.

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Recording part variations:- There are two Methods of Inspection to determine the Variation of a product from established standards.

(1) Inspection by Variables:- Here, the part is checked with a graduated tool or instrument to determine how much it varies from the ideal value which is specified for it. This method gives a record of the actual measurements of the product.

(2) Inspection by Attributes:- Here the product is checked with a 'go-not go' type of inspection tool to determine if it lies within the limits of variability established for it. This method gives a record of the total number of parts inspected and the number of part inspected and the number found to be defective.

CONTROL CHARTS.

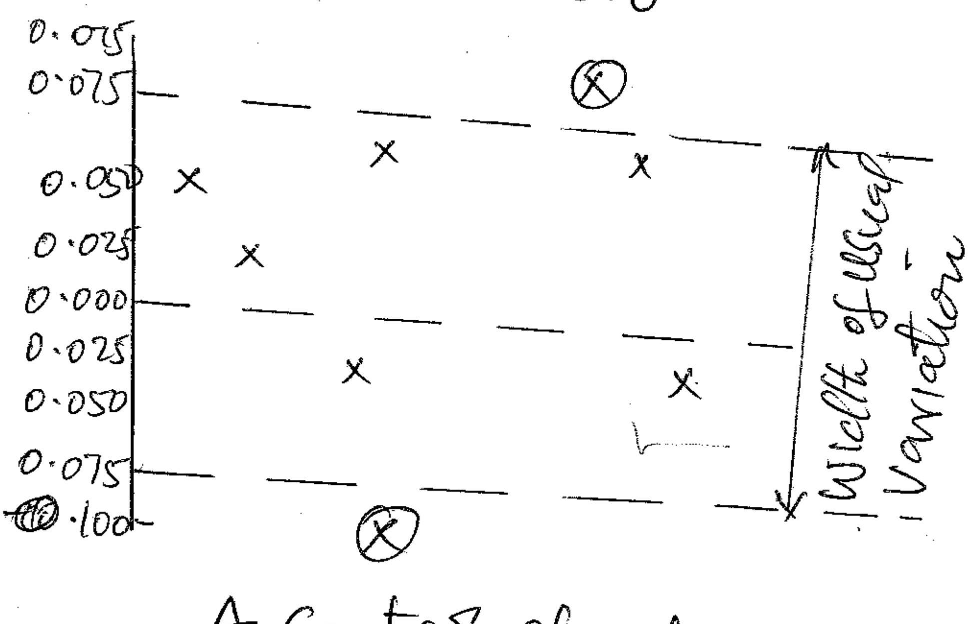
A control chart can be defined as:

A chronological (hour by hour, day by day) graphical comparison of actual product quality characteristics with limits reflecting the ability to produce as shown by past experience on the product characteristics.

The concept of the control charts can be understood by the following simple example

Two holes are to be drilled in a plate with tollerance of ± 0.075 mm between drilled hole Centres. We know that as long as the causes for Variations are only chance causes or usual causes, it will be possible to maintain these limits.

However, if assignable causes come into action the variation will be unusual, that is greater than the shopman has learned to expect (in this case ± 0.075 mm.) The operator's Immediate reaction will be that something unusual has occurred which requires corrective action. perhaps, the drill is running off-centre, perhaps, it is improperly ground, perhaps the drill jig is worn. He will try to apply possible corrective steps. This Concept is explained below.



A Control chart.

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The circled points are those requiring corrective action. The concept of usual variation limits is carried into the control charts, in the form of control limits.

TYPES OF CONTROL CHARTS.

1. Control charts for Variables:- These charts are also known as $\bar{X}$-charts and $\bar{R}$-charts since these charts gives the control limits for central tendency and spread. The basic principle for computing measurements of control chart limits is similar to that for computing the frequency distribution process, that is 3-sigma limit. 3-sigma has been chosen in place of 2 or 4-sigma, because experience has proved 3-sigma value to be most useful and economical for control charts applications.

The reason is that so many frequency distribution encountered in industry, tend towards normality and as long as only chance variation occur, there is considerable assurance that less than 3 part in 1000 would fall outside these limits.

Formulae for computing the control limits for Variables, are listed below.

(a) When Range is used in measure of spread.

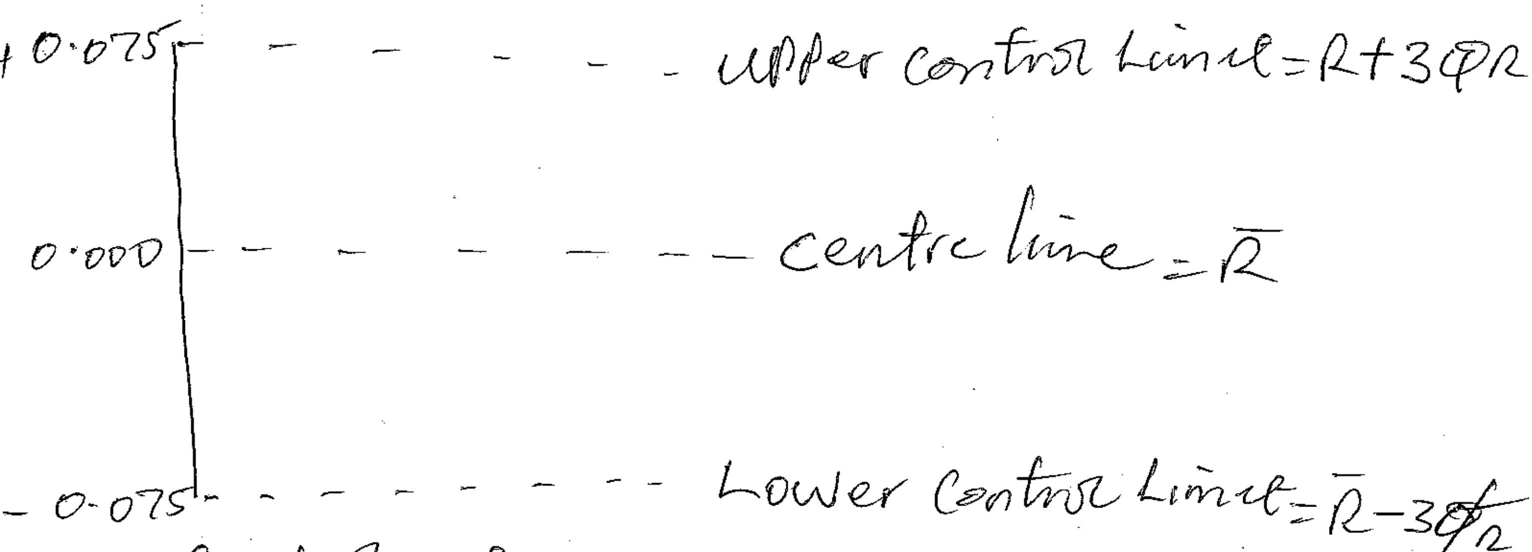
(i) Average: lower control limit (LCL) $= \bar{X} - 3\sigma$,

Centre line $= \bar{\bar{X}}$

upper control limit (UCL) $= \bar{X} + 3\sigma$.

CONTROL CHART

The concept of control chart in the case of ± 0.075mm is as below:



Control charts for Variables

Formulae for constructing control limits for variables are:

1. When Range is used in measure of spread
   - (a) Average upper ~~limit~~ control limit (UCL) = $\bar{\bar{X}} + 3\sigma$ and
   - (b) lower control limit (LCL) = $\bar{X} - 3\sigma$ while centre line = $\bar{\bar{X}}$. This is for $\bar{X}$-charts.
   - (b) Range:- $UCL = \bar{R} + 3\sigma_R$
     $LCL = \bar{R} - 3\sigma_R$
     Centre line $= \bar{R}$

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⇒ Control charts for Atributes.

The datas are usually represented in terms of fraction or Percent defective,

eg Fraction defective $p = \frac{\text{Nos of defective samples}}{n\text{(sample sizes)}}$.

Percent defective $p = p \times 100$

NB For Sub group average

~~Centre line~~ control limit $= \bar{\bar{X}} \pm 3\sigma_x$

where $\sigma_x = \sqrt{\frac{\sum_1^K (\bar{x} - \bar{\bar{x}})^2}{K}} \Rightarrow \sqrt{\frac{\bar{P}(1-\bar{P})}{n}}$

where K is the no of subgroups.

For the Range of value. we have

Control limits $\bar{R} \pm 3\sigma_R$

where $\sigma_R = \sqrt{\frac{\sum_1^K (R - \bar{R})^2}{K}} \Rightarrow \sqrt{\frac{\bar{P}(1-\bar{P})}{n}}$

NB The value 3 in the equation stands for 3-sigma control limits

(b) Range $LCL = \bar{R} - 3\sigma_R$

Centre line $= \bar{R}$

$UCL = \bar{R} + 3\sigma_R$

(2) Control Charts for Attributes:- These charts (p-charts) are used when parts are inspected by 'go' and 'not go' gauge and a unit is classified simply as good or bad. The data is represented in terms of fraction or percent defective

Fraction defective $p = \frac{\text{Number of defective in a sample}}{n \text{ (sample size)}}$.

Percent defective $p = p \times 100$

Thus if sample size is 150 and if 6 units are found to be defective, then

$$p = \frac{6}{150} = 0.04$$

and percentage defective $p = 0.04 \times 100$
$= \underline{\underline{4}}$

THE FUNCTIONS OF CONTROL CHARTS.

(a) Indicate whether the process is under control or not?

(b) Detect and determine the extent of variations is a process from the established standard.

(c) Ensure product quality

(d) Provide Information regarding the proper selection of a process.

Corresponding to the two types of Inspection Methods, There are two basic types of Control charts.

1. When the Method of Inspection is by Variables, the most popular control chart are $\bar{X}$ and R-charts

2. When the method of Inspection is by attributes, The most popular control charts are fraction or percent defective charts (also called p-charts)

To prepare the above two types of control charts the approach is similar and it involves the following steps.

1. Select the appropriate quality characteristic to be studied
2. Record data, taken on a required number of samples (usually to 10 in number) each composed of an adequate number of parts.
3. Determine the control limits from these sample data
4. check if these control limit are economically satisfactory for the job. Are they too wide? Are they too narrow?
5. Plot the control limit on suitable graph papers start to record the results of production samples of proper size and at periodic intervals.
6. Take corrective action if the characteristics of the production samples exceed the control limits.

$\bar{X}$-chart and R-chart:

These charts refer to outside diameters of shafts. In the $\bar{X}$-chart, average values of sample are plotted. Instead of individual values, because sample average tend to be more ~~nomally~~ normally distributed than single value. In the $\bar{X}$-chart, each point refers to the average of measured values (diameters) of five shafts. The centre line of $\bar{X}$-chart represents the grand average of the subgroup average $\bar{\bar{X}}$.

Control limits $= \bar{\bar{X}} \pm 3\sigma_x$

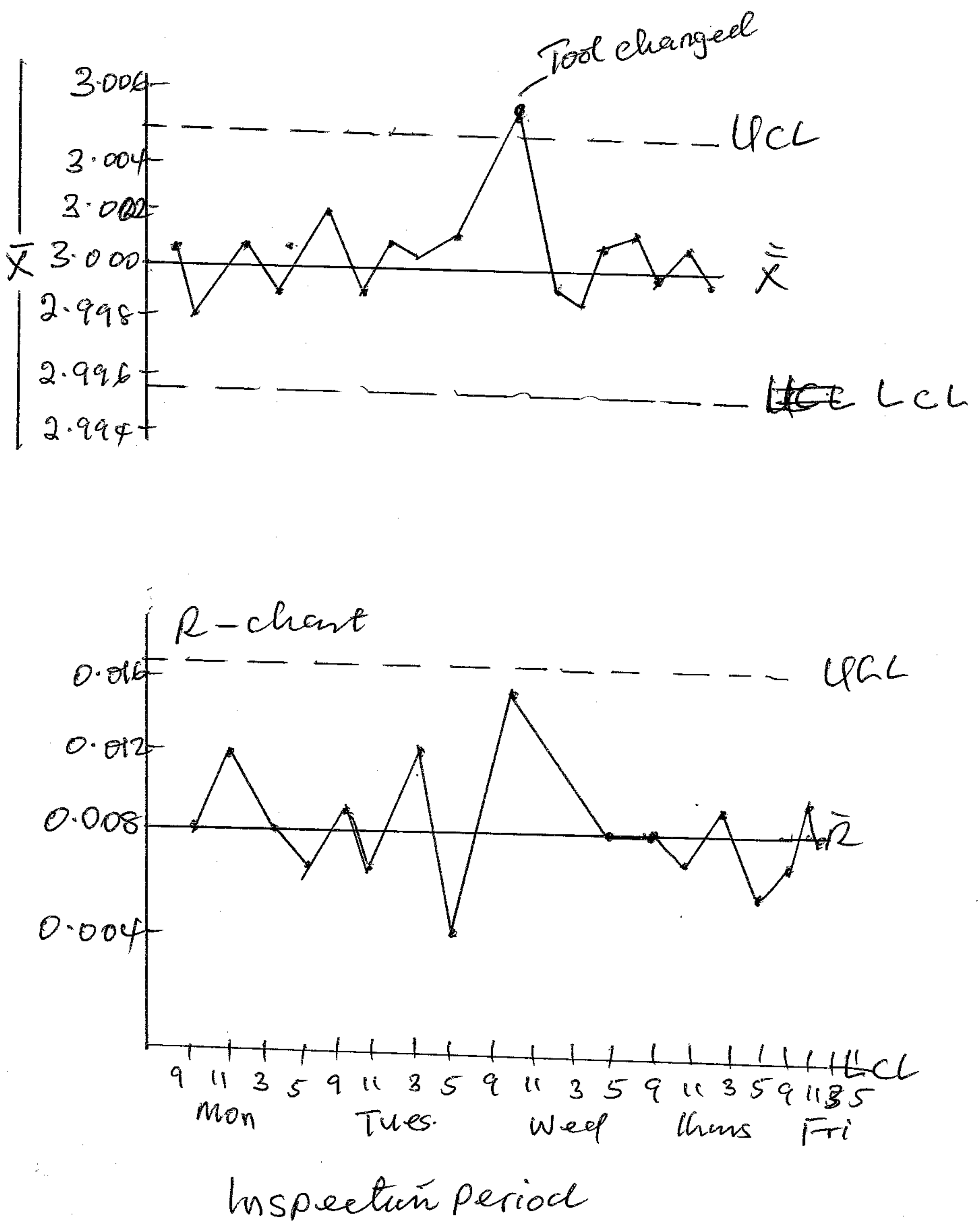
where $\sigma_x = \sqrt{\dfrac{\sum_{1}^{K}(\bar{X}-\bar{\bar{X}})^2}{K}}$

Here K is the number of subgroups.

The range of value of the R-chart are obtained from the same subgroups of 5 samples each as were in the $\bar{X}$-chart. the centre line $\bar{R}$ represent the average range of the subgroups ranges, and

Control limits $= \bar{R} \pm 3\sigma_R$

where $\sigma_R = \sqrt{\dfrac{\sum_{1}^{K}(R-\bar{R})^2}{K}}$



The fig above shows $\bar{X}$ and R-charts

Example: Plot the preliminary data with 3-Sigma control limits. If some point fall outside the limits, eliminate these point and determine the revised 3-Sigma control limits. This is explained with the help of the data given in the table below which shows the number of defectives found in daily samples of 300, for 24 consecutive production days.

Day	No of defective	Fraction defective	Day	No of defective	Fraction defective
1	15	0.05	13	12	0.04
2	8	0.027	14	21	0.07
3	15	0.05	15	6	0.02
4	18	0.06	16	15	0.05
5	17	0.057	17	17	0.057
6	14	0.047	18	17	0.057
7	33	0.11	19	39	0.13
8	6	0.02	20	20	0.067
9	18	0.06	21	15	0.05
10	36	0.12	22	14	0.047
11	32	0.107	23	17	0.057
12	23	0.077	24	18	0.06

P.T.O

Soln

Total number of Parts Sampled $= 24 \times 300$
$= 7200$

Total number of defectives found $= 446$

$\therefore \bar{P} = \frac{446}{7200} = 0.062$

$\sigma_{\bar{p}} = \sqrt{\frac{\bar{P}(1-\bar{P})}{n}}$

$= \sqrt{\frac{0.062 \times 0.938}{300}} = 0.014$

$\therefore UCL = \bar{P} + 3\sigma_p$

$= 0.062 + 3 \times 0.014$

$= 0.062 + 0.042$

$= 0.104$

$LCL = \bar{P} - 3\sigma_{\bar{p}}$

$= 0.062 - 0.042$

$= 0.020$

It is clear from the above table, that the data for days 7, 10, 11 and 19 fall outside UCL

$\therefore$ Revised Control limit will be set as below.

Now Total number of Parts Sampled $= 20 \times 300$
$= 6000$

Total number of defective $= 446 - 33 + 36 + 32 + 39$
$= 306$

Pig

$$\bar{p} = \frac{306}{6000} = 0.051$$

$$\sigma_{\bar{p}} = \sqrt{\frac{0.051 \times 0.949}{300}} = 0.013$$

Revised $UCL = 0.051 + 0.039 = 0.09$

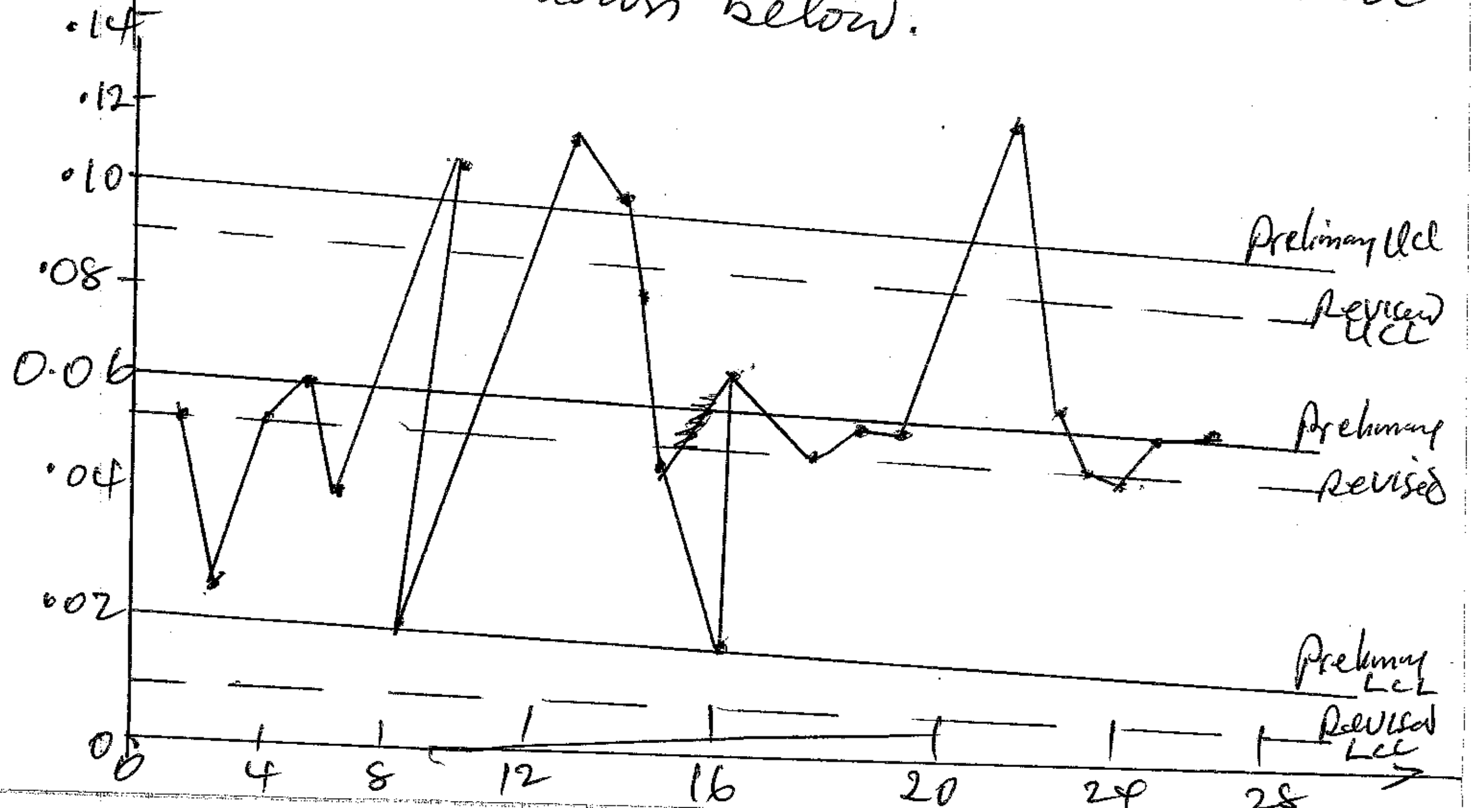
$LCL = 0.051 - 0.039 = 0.012$

These revised limits reflect the variation due to change causes.

These are used as standards for checking the future samples.

If any samples falls outside these limits, the cause will be assignable. It will be determined and corrected.

The plot of preliminary and revised control limits are as shown below.



R5

Example: In the testing of piston pins, 10 samples were taken each of 100 (sample size). The number of defectives found were respectively 3, 2, 3, 5, 3, 3, 2, 4, 3 and 2.

ⓐ Construct the control chart for fraction defectives

ⓑ Construct the percent defect control charts.

Soln

ⓐ The total sum of piston pins $= \sum p$
$= 30$

$\therefore \bar{p} = \dfrac{30}{10 \times 1000} \Rightarrow 0.03$

Now $\sigma_{\bar{p}} = \sqrt{\dfrac{\bar{p}(1-\bar{p})}{n}}$

$= \sqrt{\dfrac{0.03(1-0.03)}{100}}$

$\sqrt{\dfrac{0.03 \times 0.9\cancel{7}}{100}} = 0.017$

$\therefore$ Control limits are

$UCL = \bar{p} + 3\sigma_{\bar{p}}$

$= 0.03 + 3 \times 0.0\cancel{1}7$

$= 0.081$

$LCL = \bar{p} - 3\sigma_{\bar{p}}$

$= 0.03 - 3 \times 0.017$

$= -0.0219$

Since the negative fraction defective is meaningless $= 0$

(b) Now percent defective (Mean)

$$\bar{P} = 100 \times \bar{p}$$

$$\bar{P} = 100 \times 0.03$$

$$= 3$$

$$\sigma_{\bar{P}} = \sqrt{\frac{\bar{P}(100-\bar{P})}{n}}$$

$$\sqrt{\frac{3 \times 97}{100}} = 1.7$$

$\therefore$ Control limits are.

$$UCL = \bar{P} + 3\sigma_{\bar{P}}$$

$$= 3 + 3 \times 1.7$$

$$= 3 + 5.1$$

$$= 8.1$$

$$LCL = \bar{P} - 3\sigma_{\bar{P}}$$

$$= 3 - 3 \times 1.7$$

$$= 3 - 5.1$$

$$= -2.1 = 0$$

Centre line $= \bar{P} = 3$

With this, the Control chart can be Constructed. Since all the points lie within limits the process is under control.

Example: The data below shows the number of defective over period of 20 days in a fixed sample size of 200. Determine
(i) Whether the data exhibit statistical control.
ii Evaluate the preliminary and revised control limit for the process.

Days	No of Defectives	Fraction Defective	Day	No of Defective	Fraction Defective
1	10	0.050	11	21	0.105
2	15 18	0.075	12	15	0.075
3	10 15	0.050	13	8	0.040
4	12	0.060	14	14	0.070
5	11	0.055	15	4	0.020
6	9	0.045	16	10	0.050
7	22	0.110	17	11	0.055
8	4	0.020	18	11 13	0.055
9	12	0.060	19	26	0.130
10	24	0.120	20	13	0.065

Soln Average fraction defective $= \dfrac{\text{Total no of defective}}{\Sigma n}$

$\therefore \bar{P} = \dfrac{262}{200 \times 20} = 0.0655$

$\sigma_p = \sqrt{\dfrac{\bar{P}(1-\bar{P})}{n}} \Rightarrow \sqrt{\dfrac{0.0655(1-0.0655)}{200}}$

$$\sqrt{\frac{0.0655 \times 0.9345}{200}} = 0.0175$$

$$UCL = \bar{p} + 3\sigma_{\bar{p}}$$
$$= 0.0655 + 3 \times 0.0175$$
$$= 0.0655 + 0.0525$$
$$= 0.118$$

$$LCL = \bar{p} - 3\sigma_{\bar{p}}$$
$$= 0.0655 - 3 \times 0.0175$$
$$= 0.0655 - 0.0525$$
$$= 0.013.$$

Data for days 10 and 19 are out of control.
Therefore, for revised control limits
The total number of defectives $= 262 - (24 + 26)$
$= 262 - 50$
$= 212$

$$\bar{p} = \frac{212}{200 \times 18} = 0.0588$$

$$\sigma_{\bar{p}} = \sqrt{\frac{0.0588 \times 0.9412}{200}}$$
$$= 0.0166$$

$\therefore$ UCL $= 0.0588 + 0.0498$
$= 0.109$

LCL $= 0.009$

$\therefore$ Preliminary Control limit are.

UCL $= 0.118$
LCL $= 0.013$

and Revised Control limits are

UCL $= 0.109$
LCL $= 0.009$.

Example: A Set of assembles are subjected to inspection. A Set consists of 5 assembles and there are 15 ~~Set~~ Sub-groups. The inspection dats for a Particular day is given below. Find the Control limit for a C-chart.

Group NO	No of defects	Group NO	No of defects
1	77	9	45
2	64	10	77
3	75	11	59
4	93	12	54
5	45	13	84
6	61	14	40
7	49	15	92
8	65		

Solution

Total number of defects = 980

$\therefore \bar{C} = \frac{980}{15} \Rightarrow 65.33$

$$UCL = \bar{C} + 3\sqrt{\bar{C}}$$
$$= 65.33 + 3\sqrt{65.33}$$
$$= 65.33 + 3 \times 8.0827$$
$$= 65.33 + 24.25$$
$$= \underline{89.58}$$

$$LCL = \bar{C} - 3\sqrt{\bar{C}}$$
$$= 65.33 - 3\sqrt{65.33}$$
$$= 65.33 - 3 \times 8.0827$$
$$= 65.33 - 24.25$$
$$= \underline{41.08}$$

The data for group numbers 4, 14 and 15 fall outside these control limits. Therefore, for revised control limits we have:

$$\bar{C} = \frac{980 - (93 + 40 + 92)}{12}$$
$$= \frac{980 - 225}{12}$$
$$= \frac{755}{12}$$
$$= 62.917$$

Revised UCL $= \bar{C} + 3\sqrt{\bar{C}}$

$= 62.917 + 3\sqrt{62.917}$

$= 62.917 + 3 \times 7.932$

$= 62.917 + 23.796$

$= 86.713$

Revised LCL $= \bar{C} - 3\sqrt{\bar{C}}$

$= 62.917 - 3\sqrt{62.917}$

$= 62.917 - 3 \times 7.932$

$= 62.917 - 23.796$

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GAUGES AND GAUGES DESIGN.

INTRODUCTION:

There are several methods available for the control of dimensions of components in a system of limits and fits. Each component, for example, be measured with an instrument giving a suitable accuracy and this method is often adopted particularly for closely limited work. The method used for the majority of the work in quantity production is the system of limit gauging. This has the advantages that it can be operated in many cases by quite unskilled persons.

Gauges are inspection tools of rigid design, without a scale, which serve to check the dimension of manufactured parts. Gauges do not indicate the actual value of the inspected dimensions of the component. They are only used for determining whether the inspected part has been made within the specified limits.

A workman checking a component with a gauge does not have to make any calculations or to determine the actual

P,

dimension of the part. Gauges are easy to employ. This is one reason for their wide application in engineering. Gauges differ from measuring instrument in the following respects:-

(i) no adjustment is necessary in their use

ii They usually are not general-purpose instrument, but are specially made for some particular part, which is to be produced in sufficiently large quantities.

Gauging is used in preference to measuring when quantities are sufficiently high, because it is faster and easier with resulting lower costs.

Plain Gauges.

Plain gauges are used in checking plain, that is unthreaded holes and shafts.

⇒ Classification of plain Gauges:

Plain gauges are classified in the following ways:

(i) According to type
(ii) According to purpose
(iii) According to form of tested surface
(iv) According to design.

P2

1. According type: (a) Standard Gauges (b) Limit Gauges.

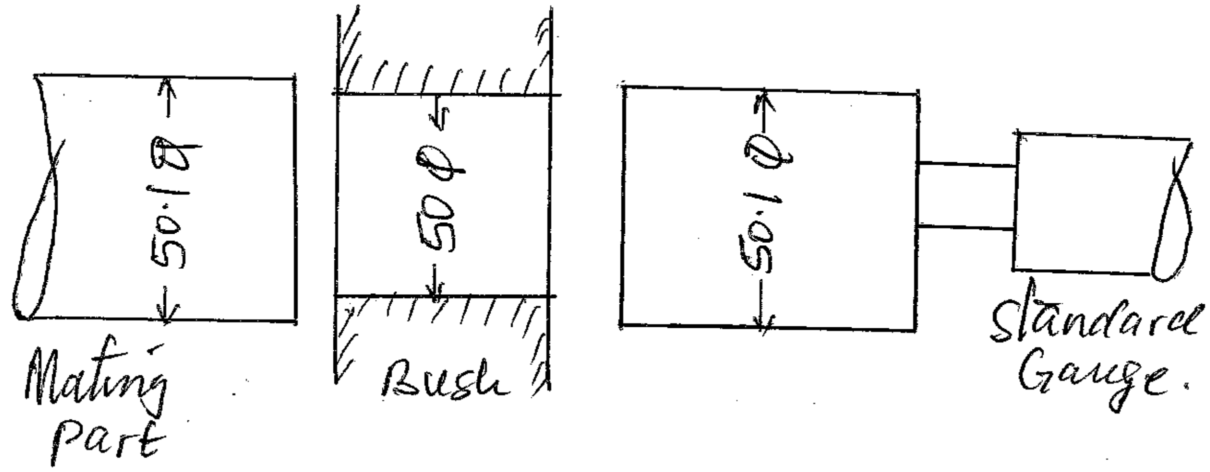
(a) Standard Gauges:- Every gauge is almost a copy of the part example a bushing is to be made which is to mate with shaft. In this case, shaft is the mating part. The bushing is ~~of the~~ checked by a gauge which in so far as the form of its surface and its size is concerned is a copy of the mating part, that is, the shaft.

If a gauge is made as an exact copy of the opposed (mating) part, in so far as the dimension to be checked are concerned, it is called a 'Standard gauge'. The first gauges to be developed were the standard gauges. The first standard gauges were the opposed (mating) parts themselves. When ~~the~~ a component is assembled with its mating parts a (mating) part itself. However, such individual fittings are not convenient or even possible in mass production conditions. Moreover, the two parts to be assembled might be in production in two different shop or even at two different plants. Therefore, it is more proper to use as a checking tool, not the mating part, but its exact copy as far as the tested dimension is concerned.

DRAWBACKS OF STANDARD GAUGE.

1. The quality of the Manufactured part will depend upon the freedom with which it ~~Means~~ Mates with the Standard gauge. The Judgement of this freedom is a relative thing and it Usually creates misunderstanding between the purchaser and the Manufacture.

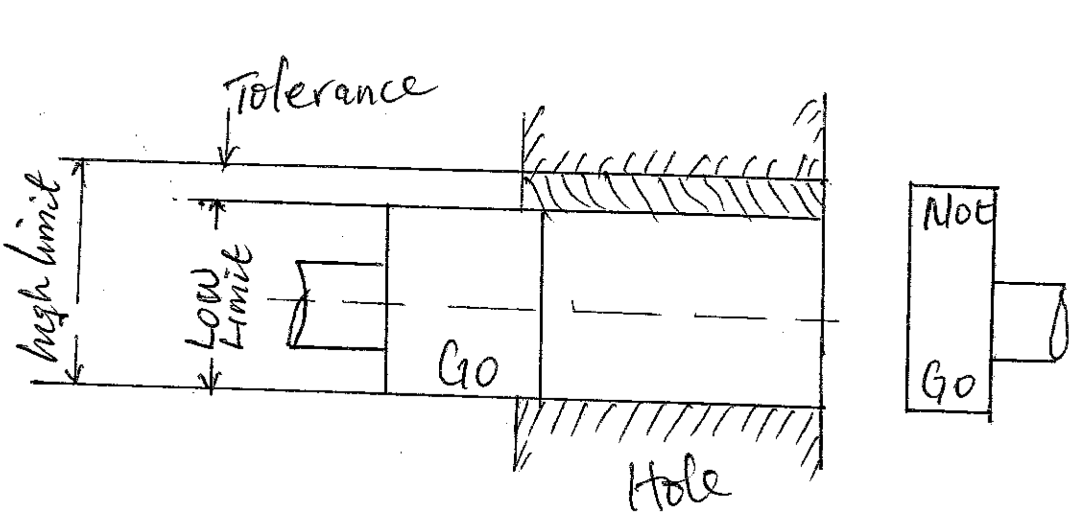
2. A Standard gauge Cannot be used to check ~~an~~ Interference fit. eg If a bushing of 50mm diameter is to be made for assembly with a shaft of 50.1mm diameter, then ~~a probe~~ the Standard gauge diameter will be 50.1mm of opposed part. Such a gauge will not pass into a properly produced bushing of diameter 50mm.



The figure above shows the application of a Standard Gauge.

Rf

(b) Limit Gauges:- The System of limit Gauges is very widely used in Industries. Limit gauges are made to the limits of the dimensions of the parts to be tested. As there are two limits of the dimensions of a part, high and low, two gauges are needed to check each dimension of the part. The part is checked by successively assembling each of the gauges with it. Since the dimensions of a properly manufactured part must be within the prescribed limits, one of the gauge called a 'Go Gauge' should pass throught or over the part, while the other gauge called a ~~over~~ 'Not Go Gauge' should not pass through or over the part. Gauge should pass through or over a part under their own weight and the part and the gauge must be at thesame temperature.



Limit Gauge

2. According to Purpose: The gauges may be classified as (a) Workshop Gauge or Working gauge (b) Inspection gauge (c) Purchase Inspection gauge and (d) Reference or Master Gauge.

(a) Workshop Gauge:- Workshop gauge or Manufacturing gauge is used by the machine operator to check the dimensions of the parts as they are being produced. These gauges usually have limits within those of the component being inspected. They are designed so as to keep the size of the part near the centre of the limit tolerance.

(b) Inspection Gauge: Inspection gauges are those used by inspectors in the final acceptance of manufactured parts when finished. These gauges are made to slightly larger tolerance than the workshop gauges so as to accept work slightly nearer the tolerance limit than the workshop gauges. This is to ensure that work which passes the working gauge will be accepted by the inspection gauge also.

(c) Purchase Inspection Gauge: The need of such gauges arises when the ~~problem~~ products of other plants are to be accepted.

The purchaser must remember that the parts may have been made and checked by working gauges worn to be the maximum permissible degree. Therefore, the 'Go' side of the purchase inspection gauge must be designed accordingly. Thus, normal size of 'Go' purchase inspection gauge will be equal to the lower limit of the hole. No-Go purchase inspection gauge design is similar to No-Go working gauge.

(d) Reference or Master Gauges:-
Reference or master gauges are used only for checking the size and condition of other gauges. Reference gauges are the reverse or opposite in form to working or inspection gauges. Due to the expenditure involved, reference gauges are seldom used and are checked by universal measuring instruments, optometers, comparators, gauge block (for snap gauge) etc.

ADVANTAGES OF LIMIT GAUGES.

1. These are conveniently used in mass production for controlling various dimensions and thus ensure interchangeability.
2. These can easily be used by semi-skilled labour.

P7

3. These are economical in their own cost as well as engaging cost.

LIMITATIONS OF LIMIT GAUGES

1. It is generally uneconomical even with excellent equipment and cost control to manufacture limit plug gauge to a tolerance on the gauge finer than 0.0013mm corresponding to a work tolerance of about 0.013mm.

2. There is also difficulty in the use of such fine limit gauges in the production shops. Finer tolerances than those specified above should preferably be measured directly with instrumental.

3. Limit Gauges only indicate whether the component is within the tolerance limit or not. They do not indicate the exact size of the component.

COMPARISON OF INDICATING INSTRUMENT GAUGES WITH LIMIT GAUGES.

- It has been found essential to use indicating instruments on 6th and finer quality components; as limit gauges can not handle the same.

- The cost of investment on indicating gauges will be recovered in the form of saving in maintenance and reconditioning costs.

The indicating instruments have the following additional benefits over limit gauges.

INDICATING GAUGES	LIMIT GAUGES
1. Indicate the actual size of the component	1. Do not indicate actual gauge.
2. Free from wear, expansion or collapse (as it in calibrated before use)	2. Susceptible for collapse, expansion and wear.
3. Require less space as a small number of gauges is sufficient.	3. Large number of gauges is required necessitating larger space.
4. Can handle 6th and finer quality components	4. Can not handle finer quality jobs
5. Require ~~ocn~~ occasional checking.	5. Require frequent checking.

CARE FOR GAUGES

Gauges should be used and cared for properly to ensure their maximum useful service life. Some suggestion for their use and care are:

1. Master, inspection and working gauges should be applied only to the uses ~~of~~ for which they are intended.

That is, a Master gauge shall be used to check inspection and working gauges, an inspection gauges will be used to the finished product and a working gauge should be used to check the product as it is being Manufactured.

(2) A plain Cylindrical gauges should be cleaned and a thin film of light oil should be applied to the gauging surface before it is used. The work should also be cleaned. Then the gauge should be aligned with the hole to be measured and give a forward motion combined with a slight rotation. The 'go plug' gauge will enter the hole if the latter is of correct size, but if not so, then the gauge will not enter it. Keep the gauge moving into and out of the hole when the fit is closed to prevent seizure of the parts.

3. Don't force a snap gauge over work because forcing will cause the gauge to pass over-sized parts and it may also spring the frame of the gauge. In fact force should be avoided in any gauging operations as it tends to harm the gauge, the work or both.

4. A Gauge should be cleaned after use and prepared for storage. If it is to be stored for a short time only, it should be coated

PTO

with a rust preventive oil If it is to be stored for a long period of time however, it should be dipped in a molten plastic material design- as a protection coating for tools and gauges.

THREAD OR SCREW GAUGES.

These gauges are designed along the same general principles as all other types of gauges. These gauges are of plug and Ring type and are made in the 'GO' and 'NOT GO' Models. These are of special quality gauge steel, hardened and seasoned before the threads are ground and finally lapped to dimensions. Nuts or internal threads are checked with plug thread gauges and screws or external threads, with ring thread gauges.

Thread gauges similar to plain gauges are designed with manufacturing tolerance for new gauges and wear allowance. Both tapered and trilock methods securing the plug gauges into their holders are employed. In case of 'GO' and 'NOT GO' plug gauges, it is common practice to make the wearing side, ie the GO end at least twice as long as the NOT GO end. The maximum wear in the latter case occurs on the end thread or threads

In screw threads, there are three classes of fit,
(a) close fit (b) Medium fit (c) Free fit.
Tolerance for Major, Minor and effective or Pitch diameters for all these three types of fits are given in national standards. Screw or thread gauges take the form of the mating thread and are assembled with the thread to be checked. By suitable designing the gauges, it is possible to provide a limit gauging system which will control the complex dimensions of the threads within the tolerance limit.

Plug screw Gauges:- Four types of plug screw gauges are used for checking the internal threads.

(a) Full form 'GO' gauge for controlling maximum metal condition of major and effective diameters simultaneously.

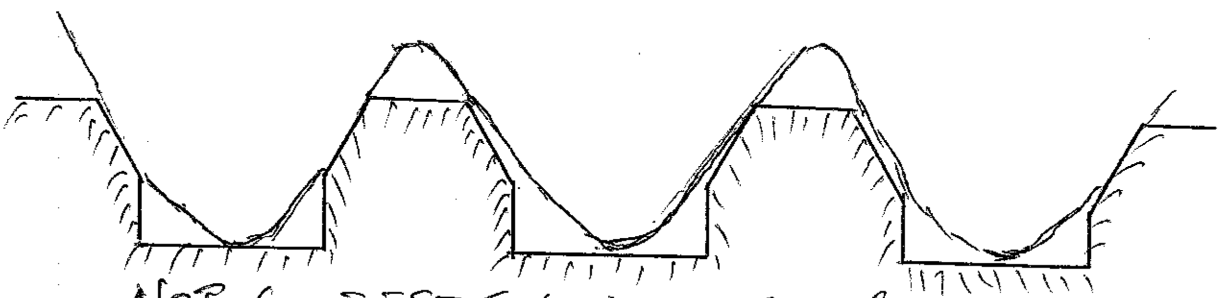
(b) 'NOT GO' effective gauge with crest and roots removed.

(c) 'NOT GO' gauge for major diameter

(d) 'NOT GO' gauge for minor diameter.

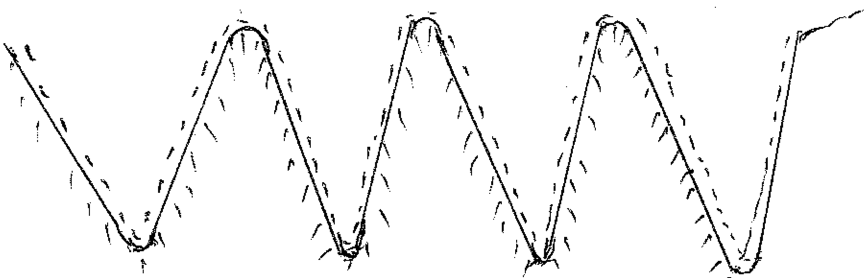
(a) Full form Go Gauge: This is made accurately to the minimum dimensions of the internal thread and will ensure that all dimensions are not less than the minimum. If it assembles with the thread, major, minor and effective diameters are checked simultaneously.

(b) 'NOT GO' effective diameter gauge:- It is a thread gauge whose diameter is truncated and whose minor diameter is cut away to be clear of the minor diameter of the work to be checked. The effective diameter of this gauge is made on the upper limit of the effective diameter of thread to be checked. This gauge is ~~of~~ very extensively used in production work.



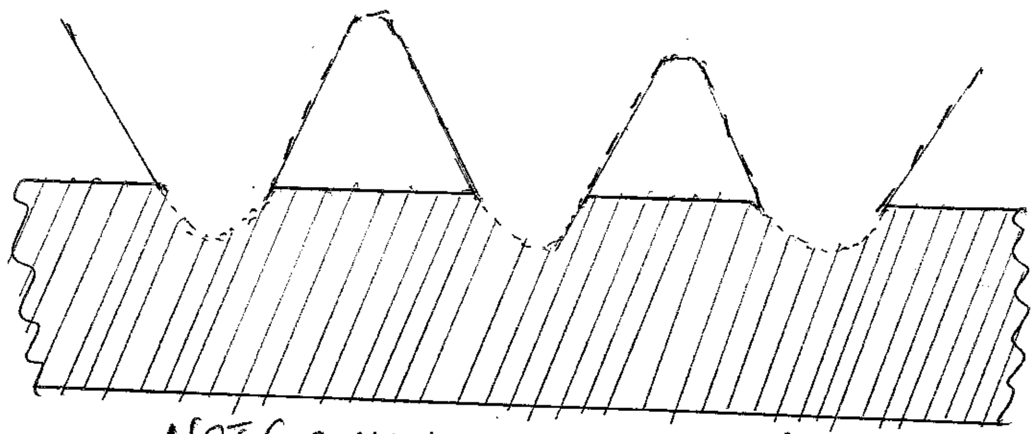
NOT GO EFFECTIVE DIAMETER GAUGE.

(c) 'NOT GO' Major diameter gauge:- This gauge has its major diameter on the corresponding upper limit of the work. Its flanks and minor diameter are well clear of the work dimension. This gauge is used to ensure that the threads of the work are not thin even though major or minor diameter is within its limits. This is not much used in production work.



NOT GO MAJOR DIAMETER GAUGE.

(d) 'NOT Go' Minor Diameter Gauge:- This gauge is used to check the core diameter only. It has its minor diameter on the corresponding upper limit of the work. This is also not in much use.



NOT GO MINOR DIAMETER GAUGE.

Plug screw gauges are usually made double ended, either solid with the 'Go' and 'NOT Go' thread gauges ground ~~so~~ on it or as a built up gauge with a metal or plastic handle ~~bar~~ having a grooved gauging unit at each end fitting into a taper socket or otherwise attached. In use a NOT Go gauge may enter the work by one or two turns and still be regarded as a NOT Go gauge.

(2) Ring Screw Gauges:- These gauges are used for checking external threads. Here also, the same ~~fore~~ four types of gauges are used. But the first two are very common. The NOT Go, effective diameter ring gauge is truncated on its minor diameter and cleared on its major diameter. Other features are the same.

Thread calliper Gauges:- These days, Instead of popular ring gauges, thread calliper gauges are much favoured. In principle, these gauges are the equivalent of gap gauges with thread forms on the anvils.

Example: Shafts of 75 ± 0.02 mm diameter are to be checked by the help of a GO, NOT GO Snap gauges. Design the gauge, sketch it and show its GO size and NOT GO size dimension. Assume normal wear allowance and gauge Marker's tolerance.

<u>Soln</u>.

High limit of Shaft $= 75.02$ mm

Low limit of Shaft $= 74.98$ mm.

Work tolerance $= 75.02 - 74.98 = 0.04$ mm

$\therefore$ Gauge Marker's tolerance (10%) $= 0.004$ mm.

Wear tolerance $= 0.002$

GO Side of Snap gauge $=$ H.L of Shaft (MMC) $= 75.02$ mm

NOT GO Side of Snap gauge $= 74.98$ mm

Wear allowance is to applied first to 'GO' side before gauge Marker's tolerance is applied.

GO side of Snap gauge after Considering the Wear allowance is $75.02 - 0.002 = 75.018$ mm

$\therefore$ Dimensions of Snap gauge are given as

P.T.O

Unilateral System

Go $75.018\text{mm}^{+0.000}_{-0.004}$ NOTGO $74.98\text{mm}^{+0.004}_{-0.000}$

Bilateral System

Go $75.018\text{mm}^{+0.002}_{-0.002}$ NOTGo $74.98\text{mm}^{+0.002}_{-0.002}$

Example:

Find the Go and NOTGo gauge dimensions of a plug gauge using Bilateral and Unilateral systems and including wear allowance for gauging 75 ± 0.05mm diameter holes.

Soln.

High limit of hole $= 75.05$mm

Low limit of hole $= 74.95$mm

Work tolerance $= 75.05 - 74.95 = 0.1$mm

Gauge makers tolerance $= 0.01$mm

Wear allowance $= 0.005$mm

Go side of plug gauge = L.L of hole (MMC)

Go side of plug gauge $= 74.95$mm.

Go side of plug gauge after the application of wear allowance $= 74.95 + 0.005$

$= 74.955$mm

Dimensions of plug gauge are given as

Unilateral System.

Go $74.955^{+0.010}_{-0.000}$ mm

NOT Go $75.05^{+0.000}_{-0.010}$ mm

Bilateral System

Go $74.955^{+0.005}_{-0.005}$ mm

NOT Go $75.05^{+0.005}_{-0.005}$ mm.